Risk Parity:
Methods and Measures of Success

WHITEPAPER

ABSTRACT

The risk parity asset allocation strategy seeks to build regime-agnostic portfolios designed to achieve financial objectives regardless of the trajectory of future economic environments. The design and subsequent evaluation of quantitative risk parity portfolios often emphasizes historical performance statistics, like Sharpe ratio and drawdowns. While these qualities feature into design and selection decisions, they are insufficient to evaluate how well a given risk parity method may respond to future market scenarios. This is especially true over the limited histories of live strategies, where samples conditioned on different regimes are profoundly inadequate to statistical inference.

We examine 14 quantitative risk parity methods and an ensemble of methods applied to a comprehensive universe of global futures markets from 1985 through May 2020. In addition to standard performance metrics, we analyze each method with a variety of diversity metrics and observe the stability of performance across each of four economic regimes. We also extend the concept of risk parity to include macro-economic factors applied to the same futures universe, with compelling results.

WHAT IS RISK PARITY, REALLY?

Most professional allocators are now familiar with the general concept of risk parity. Fundamentally, it is an asset allocation strategy designed to express the following views:

1. The future is unknowable
2. Diversification is the best defense against ignorance

These views presume:

- Diversification = Diversity + Balance
  - Diversity means owning investments with uncorrelated fundamental responses to different economic environments
  - Balance means striving to allow every investment to have an equal expected impact on the portfolio's return and risk
- A diversified portfolio should maximize exposure to a diverse set of systematic risks, while minimizing exposure to idiosyncratic risks
  - Just as there are three primary colors, there are a finite number of fundamental sources of risk, for which investors may expect to earn excess returns
  - On average, these risks should produce an equal amount of excess return per unit of risk
  - All other risk is unrewarded and should be diversified away

In short, a risk parity portfolio should harness the power of diversification to make the most of whatever the future holds.

MULTIPLE PERSONALITY DISORDER

While risk parity may be rather simple in concept, as with most things there are devils in the details.

For example, there is little consensus among practitioners about how to define - and perhaps more importantly measure - fundamental inputs like diversity; balance; economic environments; and systematic risk factors.

Should we measure diversity in terms of how well our investment universe spans exposure to demand shocks? Supply shocks? Inflation? Liquidity? Or are the true sources of risk un-observable and endogenous to markets themselves? Are we concerned about deviations in growth and/or inflation rates between countries or regions? Or are the true sources of risk un-observable and endogenous to markets themselves?

How can we measure balance until we define the risks that we are trying to create balanced exposures to?

These questions are further complicated by the fact that, while a risk parity portfolio is designed to earn excess returns from a diverse set of risks, those risks may not occur with equal frequency. How should we think about diversity and balance if one regime is expected to manifest just 5 percent of the time with an extremely large impact, while another regime is expected to occur half the time, but with more moderate impact?

Moreover, how should we measure the success of a risk parity strategy? Should we rely on how well the strategy performed on average over a certain period of time? Or should we judge the merits of a risk parity approach based on how well it managed to deliver balanced exposure to target risks, notwithstanding average results?

The fact is, investors seek to answer these questions in a variety of ways. They identify different risks for which they expect to be compensated, and they seek diversity and balance relative to these target risks. Each framework for thinking about risk and diversity prompts a unique method for constructing portfolios, so risk parity can take on many different personalities to suit unique objectives.

In the end however, most reasonable investors want solutions that maximize expected risk-adjusted returns while minimizing vulnerability to any particular type of market environment. This is the ultimate objective for risk parity portfolios, but this objective is not explicitly expressed by any of the prevailing methods! Rather, investors approximate this objective by attempting to balance risk across markets, or asset classes, or clusters, or directions of market risk. But it is critical to remember that these objectives are only proxies for the true economic objectives of investors.

THE RISK PARITY ECOSYSTEM

Risk parity portfolios prioritize diversification across the widest possible investment universe. In addition, it is often desirable to apply leverage to diversified risk parity portfolios to achieve higher return targets. For these and other reasons (i.e. liquidity, central clearing, fungibility, etc.) many professional investors use futures markets as their fundamental unit of exposure. Futures markets provide leveraged exposure to individual securities (such as barrels of physical oil, or a notional government bond) as well as aggregates of securities (such as the S&P 500 equity index).

Some risk parity implementations view the concept from the perspective of maximizing diversification across all of the available markets, i.e. the markets described in Figure 1. Inverse volatility, inverse variance, and equal risk contribution weighting methods seek to divide nominal risk equally across all markets, but with different assumptions about correlations and relationships between risk and return. These methods emphasize diversity by treating all markets equally, but may introduce risk biases if the investment universe is out of balance[1].

Perhaps the most straightforward way to think about risk and diversity is at the asset class level. If we espouse the premise that broad asset classes like stocks, bonds and commodities represent unique risk factors, which would be expected to produce excess returns in different economic environments, then a risk parity portfolio would seek to allocate equal risk to each asset class cluster.

There are several ways to achieve this objective. For example, one could simply cluster markets heuristically into stock indices, bond indices, and commodities. Alternatively, cluster structures can emerge quantitatively from correlations or other distance metrics derived from underlying returns. The Hierarchical Risk Parity method introduced by Lopez de Prado[2] is an example of the latter method. Markets are then risk-weighted within their clusters, and then risk-weighted across clusters.

The methods above are suitable for risk parity frameworks where systematic risk is aligned in the directions described by markets and asset classes. However, there are other approaches that attempt to look through market returns to find the latent sources of risk that emerge from common features across markets and asset classes.

For example, Principal Component Analysis (PCA) can be used to isolate the independent sources of risk across a universe of markets. Each principal component can be thought of as a principal portfolio (PP), which is made up of long and short positions in each market. For example, the first principal portfolio may have positive weights in equity indices, but also small negative weights in bonds, since some of the variance in bond returns may be explained by negative equity beta. Each principal portfolio expresses a unique source of risk that is, by construction, uncorrelated to all other sources of risk.

Importantly, while PCA produces an array of factors with dimensions equal to the number of variables (i.e. markets), most of these factors have very small explanatory power and may be thought of as random noise (i.e. idiosyncratic risk). Obviously, as portfolio managers we are interested in gaining exposure to economically relevant sources of risk and return, while minimizing exposure to idiosyncratic risk. So our objective is to identify relevant factors while discarding insignificant ones. This is called “eigenvalue clipping.”

[1] For example, if our universe consisted of many equity-like instruments with just a single bond instrument

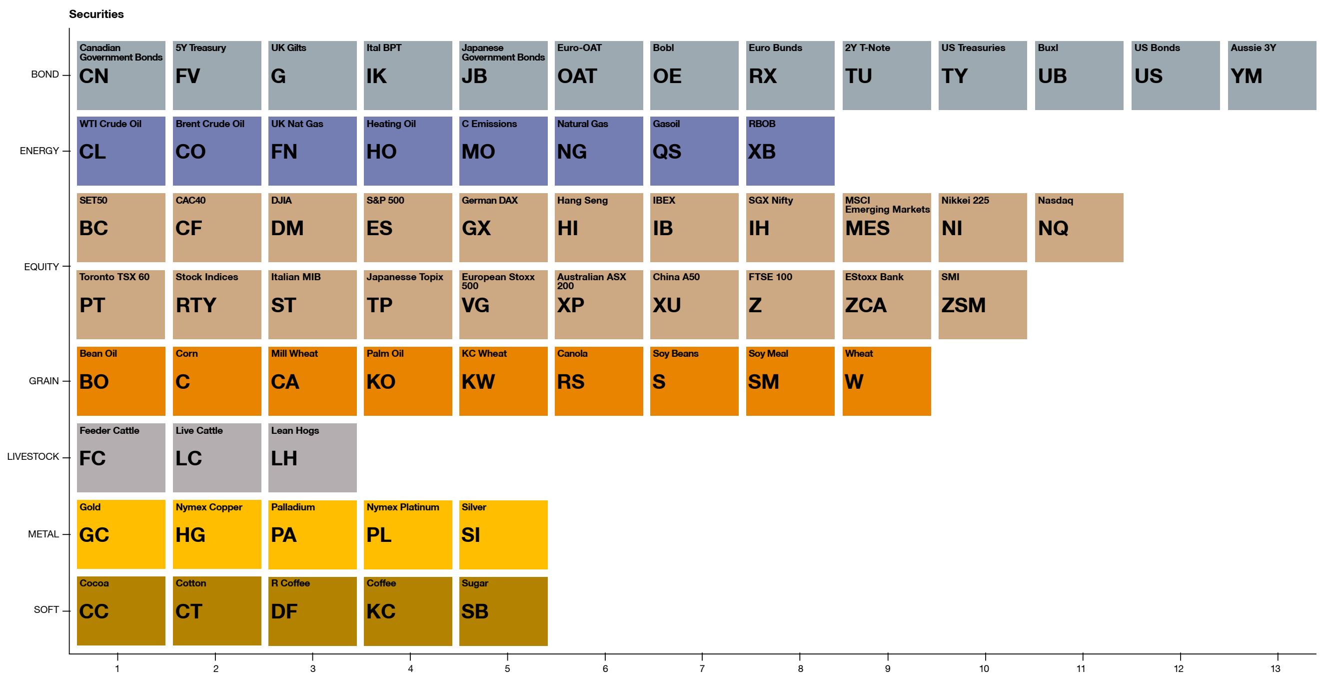
[2] See Building Diversified Portfolios that Outperform Out-of-Sample

There are many ways to identify significant factors (and other ways to preserve all factors, but adjust the eigenvalues). One approach is to find the eigenvalue spectrum (i.e. variance explained by each principal component) that would be expected from PCA on randomly formed correlation matrices. If a principal portfolio explains less variance than might be expected from random matrices, then it is rejected as insignificant.

PCA is analogous to multi-variate regression or portfolio optimization, and suffers from the same instabilities. There are a variety of ways to manage these instabilities, but perhaps the simplest and most robust approach is to impose non-negativity constraints on the factor loadings. This is analogous to imposing long-only weights for portfolio optimization, and may have the added benefit of more intuitively mapping principal portfolios to recognizable economic clusters, like equities, bonds and commodities.

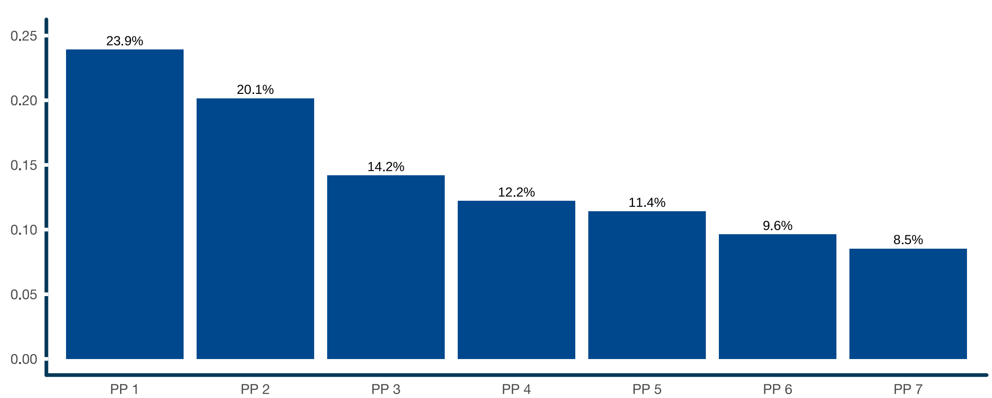
For illustration we found the significant long-only principal portfolios that emerge from the daily returns of our futures universe in calendar 2019. There are seven significant principal portfolios in this period [3].

Figure 1: **Universe of futures markets**



Source: ReSolve Asset Management. For illustrative purposes only.

Figure 2: **Proportion of standardized volatility explained by significant principal portfolios**

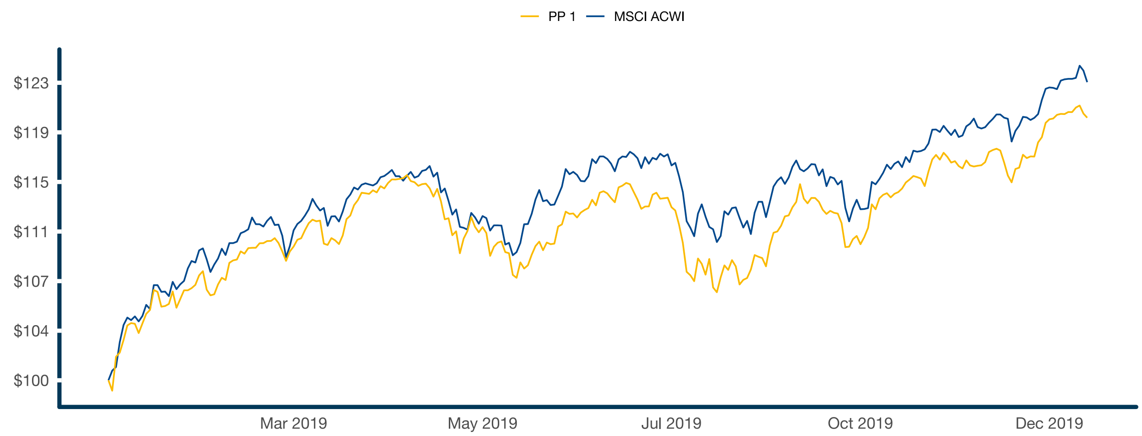


Source: Data from CSI. Analysis by ReSolve Asset Management. SIMULATED RESULTS.

3 Based on Horn's test.

It is often challenging to derive economic meaning from the factors that emerge through PCA. However, the introduction of long-only constraints increases the chances that principal portfolios will generally align with known dimensions of risk in the universe of markets under investigation. We project the in-sample growth trajectory for three principal portfolios for calendar 2019 below. By inspection, it's clear that the first principal portfolio reflects "growth" risk, since it loads mostly on growth-sensitive markets like equities, oil, and copper. The second principal portfolio loads mostly on global sovereign bond futures. Principal portfolio 5 seems to track a typical commodity index. Figure 3, Figure 4 and Figure 5 illustrate this phenomenon by plotting the performance of principal portfolios against respective asset class proxies.

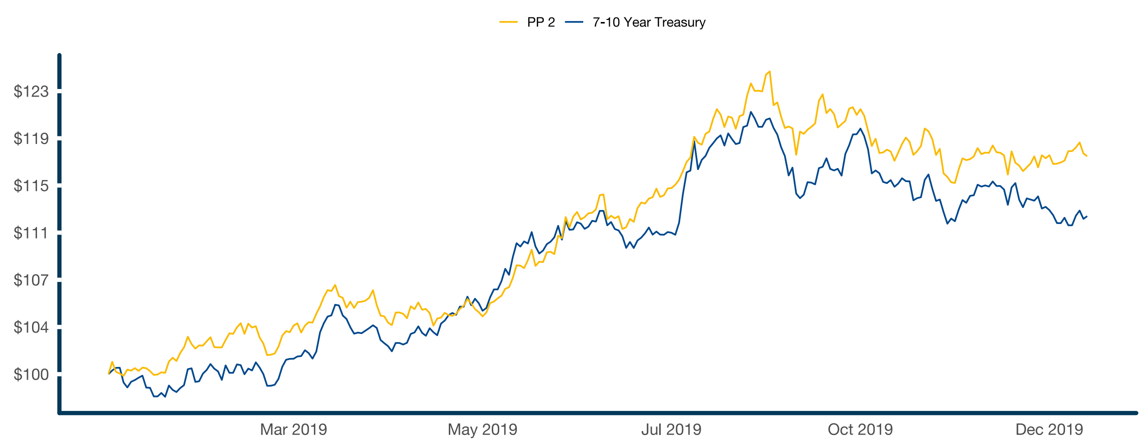
Figure 3: **PP 1 vs. iShares MSCI ACWI ETF excess returns, 2019**



THESE RESULTS ARE BASED ON SIMULATED OR HYPOTHETICAL PERFORMANCE RESULTS THAT HAVE CERTAIN INHERENT LIMITATIONS. UNLIKE THE RESULTS SHOWN IN AN ACTUAL PERFORMANCE RECORD,THESE RESULTS DO NOT REPRESENT ACTUAL TRADING. ALSO, BECAUSE THESE TRADES HAVE NOT ACTUALLY BEEN EXECUTED, THESE RESULTS MAY HAVE UNDER-OR OVER-COMPENSATED FOR THE IMPACT, IF ANY, OF CERTAIN MARKET FACTORS, SUCH AS LACK OF LIQUIDITY. SIMULATED OR HYPOTHETICAL TRADING PROGRAMS IN GENERAL ARE ALSO SUBJECT TO THE FACT THAT THEY ARE DESIGNED WITH THE BENEFIT OF HINDSIGHT. NO REPRESENTATION IS BEING MADE THAT ANY ACCOUNT WILL OR IS LIKELY TO ACHIEVE PROFITS OR LOSSES SIMILAR TO THESE BEING SHOWN.

Source: Data from CSI. Analysis by ReSolve Asset Management. SIMULATED RESULTS.

Figure 4: **PP 2 vs. iShares 7-10 Year Treasury Index ETF excess returns, 2019**



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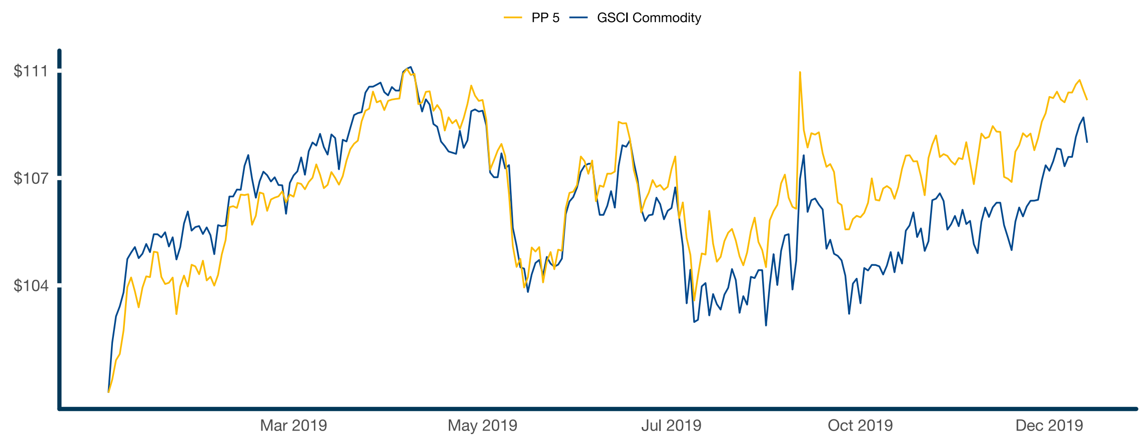
Source: Data from CSI. Analysis by ReSolve Asset Management. SIMULATED RESULTS.

Figure 5: **PP 5 vs. iShares S&P GSCI Commodity Index ETF excess returns, 2019**



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Source: Data from CSI. Analysis by ReSolve Asset Management. SIMULATED RESULTS.

Once the significant independent sources of risk have been identified, it is a simple matter to form a portfolio that distributes risk equally across the significant principal factors. This approach, which was first proposed by Meucci[4] and Lohre[5] may address some challenges with methods that rely on covariances at the market level. Several approaches have been proposed to distribute risk to latent factors, including Agnostic Risk Parity and Minimum Torsion. We explore these below.

DATA AND METHODS

Our investment universe consists of the 64 futures markets described in Figure 1: **Universe of futures markets**. We remove currency markets because there is no coherent way to define long or short exposure since one can only be long one currency by being short in another. Importantly, this universe is reasonably diversified by construction, which is to the benefit of risk parity methods that do not explicitly account for correlations in portfolio formation.

While some futures markets have data back to the early 1970s, S&P 500 and benchmark Treasury and UK Gilt futures only started trading in 1982. Aussie 200 futures launched in 1983 while FTSE 100 futures launched in 1984. Accounting for a one-year estimation window for portfolio parameters we start tracking all simulations in 1985[6]. Markets are added once they have one year of returns.

Continuous contract returns are computed by rolling front-month contracts when two consecutive days of both volume and open interest have migrated to the next contract.

[4] 2009, "Managing Diversification", Risk 22, 74–79

[5] 2012, "Diversifying Risk Parity", Journal of Risk, Vol. 16, No. 5, 2014, pp. 53-79

[6] Readers interested in exploring the long-term character of risk parity strategies may refer to Risk Parity Part II: The Long Run View and Demystifying Risk Parity Through 90 Years of History

Figure 6: **Performance characteristics of continuous contract futures contracts**

	Start Date	Annualized Return	Annualized Volatility	Sharpe Ratio	Max Drawdown	Skewness	Excess Kurtosis
SET50	2006-08-25	1.85%	13.20%	0.20	-61.60%	-0.12	10.67
Bean Oil	1974-01-03	-2.15%	24.00%	0.03	-90.70%	0.17	-1.48
Corn	1972-11-21	-4.31%	23.00%	-0.08	-96.80%	0.16	-0.4
Mill Wheat	2006-07-04	1.68%	11.00%	0.21	-56.70%	0.2	1.1
Cocoa	1978-01-04	-6.25%	26.90%	-0.11	-98.60%	0.14	-0.77
CAC40	1988-08-19	2.90%	17.50%	0.25	-67.20%	-0.15	2.11
WTI Crude Oil	1983-10-04	0.70%	32.90%	0.19	-98.40%	-0.6	23.06
Canadian Government Bonds	1989-09-18	2.29%	4.80%	0.50	-16.00%	-0.16	-0.72
Brent Crude Oil	1989-03-02	2.61%	27.60%	0.23	-92.20%	-0.58	12.22
Cotton	1972-11-15	-0.18%	23.10%	0.11	-93.40%	0.05	-1.53
R Coffee	1980-01-03	-1.51%	26.70%	0.08	-90.80%	0.58	9.49
DJIA	1997-10-07	2.39%	13.10%	0.24	-52.70%	0.13	12.27
S&P 500	1982-06-14	5.50%	16.90%	0.40	-62.90%	-1.11	45.4
Feeder Cattle	1972-07-04	1.36%	15.90%	0.16	-68.00%	-0.08	-1.35
UK Nat Gas	2010-01-05	-6.17%	16.90%	-0.29	-97.40%	0.63	3.12
5Y	1988-05-27	1.78%	3.20%	0.57	-9.50%	-0.07	0.81
UK Gilts	1982-11-19	2.12%	6.40%	0.36	-30.50%	0.03	0.68
Gold	1975-01-29	0.06%	18.10%	0.09	-92.80%	0	4.1
German DAX	1990-12-19	2.67%	17.40%	0.24	-75.60%	-0.05	3.7
Nymex Copper	1973-09-05	1.78%	25.80%	0.20	-88.70%	-0.04	0.57
Hang Seng	1991-01-03	4.92%	20.70%	0.34	-64.20%	0.55	10.51
Heating Oil	1981-01-06	0.70%	29.20%	0.17	-89.70%	-0.3	7.45
IBEX	1992-04-21	2.30%	17.50%	0.22	-60.10%	-0.22	3.71
SGX Nifty	2011-05-18	0.01%	7.60%	0.04	-40.30%	-1.01	10.71
Ital BTP	2009-09-15	1.43%	4.40%	0.34	-19.30%	0.11	10.38
Japanese Government Bonds	1990-04-05	1.97%	3.10%	0.65	-9.80%	-0.29	1.96
Coffee	1973-01-03	-3.39%	33.00%	0.06	-98.50%	0.48	5.5
Palm Oil	1992-10-13	3.30%	18.60%	0.27	-73.70%	0.06	0.68
KC Wheat	1972-08-25	-2.64%	23.80%	0.01	-93.70%	0.23	-0.18
Live Cattle	1972-07-13	1.25%	16.60%	0.16	-63.90%	-0.12	-1.07
Lean Hogs	1972-07-25	-1.29%	25.30%	0.07	-96.20%	0.02	-1.73
MSCI Emerging Markets	2010-12-16	-0.17%	9.90%	0.03	-41.50%	-0.68	6.32

	Start Date	Annualized Return	Annualized Volatility	Sharpe Ratio	Max Drawdown	Skewness	Excess Kurtosis
C Emissions	2008-01-03	-0.68%	24.90%	0.10	-92.10%	-0.18	7.41
Natural Gas	1991-10-02	-10.84%	37.50%	-0.12	-99.90%	0.25	-0.06
Nikkei 225	1986-09-04	0.09%	20.20%	0.11	-84.90%	-0.13	25.75
Nasdaq	1996-06-12	4.47%	19.80%	0.32	-84.80%	0.12	3.91
Euro-OAT	2012-04-20	0.99%	2.00%	0.50	-7.10%	-0.31	0.32
Bobl	1991-10-07	1.59%	2.40%	0.66	-8.30%	-0.26	-0.4
Palladium	1978-01-04	6.44%	29.80%	0.36	-88.20%	0.03	5.13
Nymex Platinum	1972-07-06	0.78%	25.50%	0.16	-87.20%	-0.11	0.51
Toronto TSX 60	1999-09-08	1.78%	12.50%	0.20	-56.10%	-0.43	12.3
Gasoil	1981-04-07	2.38%	28.00%	0.22	-90.00%	-0.18	7.78
Canola	1972-07-04	-0.11%	20.00%	0.10	-82.70%	0.01	-0.85
Russell 2000	1995-01-04	3.01%	16.90%	0.26	-60.40%	-0.4	4.63
Euro Bunds	1990-11-26	2.55%	4.10%	0.64	-11.10%	-0.2	-0.89
Soybeans	1972-07-04	1.63%	23.40%	0.19	-86.10%	-0.02	0.14
Sugar	1972-07-04	-3.96%	36.10%	0.07	-99.60%	0.04	-0.46
Silver	1972-07-06	-0.91%	29.20%	0.12	-98.20%	-0.32	1.18
Soy Meal	1973-01-03	3.22%	26.60%	0.25	-87.60%	0.19	1
Italian MIB	1995-03-16	0.99%	17.50%	0.14	-73.60%	-0.25	4.18
Japanese Topix	1990-04-04	-0.65%	17.80%	0.05	-76.60%	0.13	6.54
2Y	1990-06-25	0.83%	1.30%	0.67	-4.50%	0.06	4.81
US Treasuries	1982-05-04	3.51%	5.80%	0.62	-16.10%	0.11	0.78
Buxl	2005-09-13	1.94%	6.70%	0.32	-19.40%	-0.14	-0.56
US Bonds	1978-08-02	3.44%	10.30%	0.38	-46.80%	-0.02	-0.43
European Stoxx 50	1998-06-23	0.56%	16.20%	0.12	-68.50%	0.03	3.01
Wheat	1972-07-04	-5.49%	26.40%	-0.08	-99.00%	0.22	-0.75
RBOB	1985-03-04	4.62%	30.60%	0.30	-85.90%	-0.35	7.5
Australian ASX 200	1983-02-17	3.90%	16.70%	0.32	-56.20%	-2.75	76.16
ChinaA50	2010-10-22	0.69%	10.70%	0.12	-45.20%	-0.46	6.83
Aussie 3Y	1988-05-18	0.41%	1.00%	0.42	-4.80%	-0.21	2.08
FTSE 100	1984-05-04	2.47%	15.70%	0.23	-56.40%	-0.43	7.3
EStoxx Bank	2010-01-05	-1.83%	15.70%	-0.04	-66.80%	-0.32	5.68
SMI	1998-10-02	2.27%	12.60%	0.24	-57.40%	-0.07	4.75

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Source: Data from CSI. Analysis by ReSolve Asset Management. SIMULATED RESULTS.

We perform a walk-forward optimization which rebalances portfolios weekly in four tranches. The objective is to approximate a monthly rebalance horizon, but reduce sensitivity to end-of-month rebalance dates and reduce noise that might be introduced by good or bad timing luck. At each rebalance date t we use our variance-covariance matrix estimate (EWMA on daily returns with lambda=0.97) to form optimal portfolios for each risk parity method. Given these optimal portfolio weights, we use a shorter-term variance-covariance estimate (EWMA with lambda=0.9) to scale the portfolio to target 10 percent annualized volatility. We assume portfolios are rebalanced at the close of the day following portfolio formation (t+1), so that returns start accruing to the new portfolio at (t+2).

We investigate the following methods for constructing risk parity portfolios:

- INV-VOL : Inverse Volatility weighted portfolios weight markets by the inverse of their volatility. This is optimal if markets have similar expected Sharpe ratios and similar expected pairwise correlations, both within and across asset classes.
- INV-VAR : Inverse Variance weighted portfolios weight markets by the inverse of their variance. This is optimal if markets have similar expected returns (irrespective of risk), and similar expected pairwise correlations, both within and across asset classes.
- Cluster Risk Parity weighted portfolios first cluster markets based on their similarity, and then risk-weight markets first within clusters, and then across clusters. Weights might be proportional to inverse volatility or variance.
    - RP-HC : Heuristic Clustering employs human intuition to divide markets into sensible clusters, for example by asset class, region, commodity sector, etc.
    - Hierarchical Risk Parity uses hierarchical methods to sort markets into sequentially granular clusters based on a quantitative dissimilarity measure.
        - HRP-VOL : Hierarchical Risk Parity where clusters are sequentially weighted by the inverse of cluster standard deviation
        - HRP-VAR : Hierarchical Risk Parity where clusters are sequentially weighted by the inverse of cluster variance
- Equal Risk Contribution weighted portfolios ensure that each market contributes the same amount of volatility to the portfolio, after accounting for estimated covariances. This method is a convenient heuristic if the composition of asset class clusters are of roughly equal size.
- Maximum Diversification weighted portfolios seek to maximize the number of independent directions of risk that are expressed in the portfolio by executing a mean-variance optimization. As a result, the portfolio can be especially sensitive to errors in covariance estimates, and portfolios can be highly concentrated in a few markets.
- Hierarchical Sub-Space Risk Parity : Portfolios are formed by optimizing many smaller investment universes drawn randomly from the full investment universe. Markets are drawn from clusters informed by similarity to ensure each sample universe contains exposures to all major directions of risk.
    - HRSS MAX-DIV : Sub-portfolios are optimized for Maximum Diversification
    - HRSS MIN-VAR : Sub-portfolios are optimized for Minimum Variance
- Principal Component Parity methods seek to distribute risk equally across the significant uncorrelated directions of risk that emerge from the data at each rebalance period.
    - MAX BETS PCA COR : Equalize risk contributions across modified Principal Portfolios derived from the correlation matrix
    - MAX BETS PCA COV : Equalize risk contributions across modified Principal Portfolios derived from the covariance matrix
    - MIN-TORSION-COR : Equalize risk contributions across minimum torsion factors derived from the correlation matrix
    - MIN-TORSION-COV : Equalize risk contributions across minimum torsion factors derived from the covariance matrixARP : Agnostic Risk Parity seeks to create a portfolio that balances the risk-return trade-off over all principal components of asset covariance while minimizing risks generated by over-optimistic hedging of the different sources of risk via a Rotationally Invariant Portfolio[7]

We also examined an ENSEMBLE of all methods. At each rebalance we found the contemporaneous covariances between models based on current model weights and the market covariance matrix. We used these estimates to form the robust Maximum Diversification portfolio of all individual methods. The ensemble was created with a one-day delay to accommodate the extra step.

RESULTS

We start with an examination of portfolio performance over the full period from 1985 through May 2020. Aside from annualized returns and Sharpe ratios we are interested in higher moments and the duration and magnitude of drawdowns as measured by the Ulcer Ratio[8]. We are

[7] See Agnostic Risk Parity: Taming Known and Unknown Unknowns

[8] The Ulcer Index captures both the depth and the duration of drawdowns and is a good measure of how much anxiety a hypothetical investor would have endured from investing in a strategy. It is the geometric average of all instances where an investor's wealth dropped below an all-time high, and accounts for both the depth and duration of wealth impairments. $R_i = \frac{P - MAX(P)}{MAX(P)}$ $\quad Ulcer = \sqrt{\frac{R_1^2 + R_2^2 + ... + R_N^2}{N}}$

also interested in leverage ratios. All returns are excess returns gross of fees and transaction costs but net of leverage costs, since leverage costs are embedded in futures.

Figure 7: **Performance of risk parity simulations - 1985 - May 2020 - SIMULATED RESULTS**

	Excess Return	Volatility	Sharpe Ratio	Max Drawdown	Return to Ulcer	Skewness	Excess Kurtosis	Average Leverage
INV-VOL	9.23%	11.20%	0.84	-30.90%	0.95	-0.33	-0.69	3.58x
INV-VAR	12.05%	9.80%	1.21	-26.10%	1.56	-0.04	0.22	7.72x
MAX-DIV	10.90%	10.70%	1.02	-29.40%	1.22	-0.24	-0.68	5.2x
ERC	11.56%	11.20%	1.04	-31.90%	1.24	-0.32	-0.3	4.89x
HRSS MAX-DIV	11.63%	11.00%	1.06	-30.50%	1.40	-0.25	-0.74	4.13x
HRSS MIN-VAR	13.47%	10.50%	1.26	-24.10%	1.75	-0.19	-0.08	6.98x
HRP-VOL	9.68%	11.20%	0.88	-30.80%	1.01	-0.34	-0.27	3.9x
HRP-VAR	10.61%	9.40%	1.12	-27.50%	1.26	-0.13	1.76	8.13x
MAX BETS PCA COV	9.67%	9.00%	1.07	-25.20%	1.40	-0.16	0.11	7.92x
MAX BETS PCA COR	9.81%	9.20%	1.06	-23.30%	1.34	-0.15	0.21	5.59x
MIN-TORSION-COV	11.17%	10.80%	1.03	-31.60%	1.23	-0.27	-0.56	5.57x
MIN-TORSION-COR	9.66%	9.80%	0.99	-27.90%	1.19	-0.1	0.35	6.48x
ARP	10.68%	11.10%	0.97	-31.90%	1.11	-0.32	-0.45	4.62x
RP-HC	12.40%	11.30%	1.09	-30.00%	1.46	-0.42	0.89	4.7x
ENSEMBLE	11.14%	10.00%	1.11	-24.80%	1.45	-0.23	-0.64	5.55x

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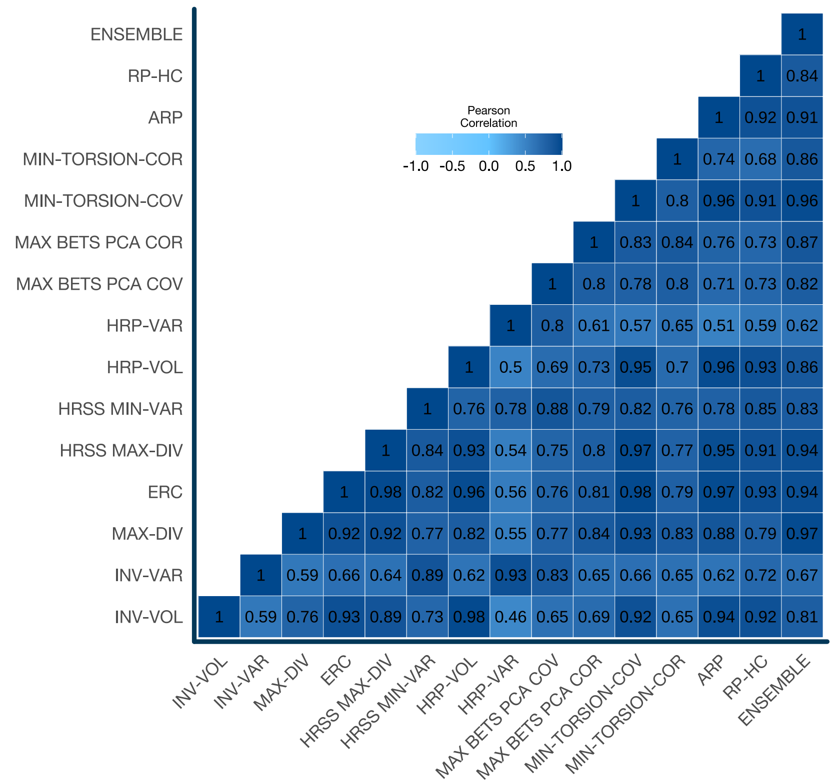
Source: Data from CSI. Analysis by ReSolve Asset Management. SIMULATED RESULTS.

All of the risk parity strategies produced attractive performance over the sample period. Most Sharpe ratios are close to 1. Strategies exhibit mild negative skewness and minimal excess kurtosis compared with individual markets, which averaged 0.28 and 6.25 (absolute values), respectively. This is compelling evidence that diversification is very effective in truncating the so-called “fat-tails” of return distributions so that they may be effectively modeled under log-normal assumptions.

Variance-weighted strategies appear to have a slight edge in the data from the perspective of both Sharpe ratio and drawdowns. However, as we will discuss below, these strategies may have other vulnerabilities that cannot be teased out from long-term performance statistics. The ENSEMBLE method compares favorably with other methods on Sharpe ratio, Maximum Drawdown, and Return-to-Ulcer ratio, reinforcing the power of model diversification. However, none of the differences in Sharpe ratios are statistically significant.

It is also interesting to observe the correlations between risk parity strategies formed on the same universe, rebalanced on the same dates, and based on the same correlation / covariance estimates. The average pairwise correlation between methods is 0.79 but they range from 0.46 at the low end to 0.98 at the upper end. It’s clear that different risk parity methods produce portfolios with surprisingly diverse qualities.

Figure 8: **Pearson correlations of daily returns for risk parity methods 1985 - May 2020 - SIMULATED RESULTS**



	INV-VOL	INV-VAR	MAX-DIV	ERC	HRSS MAX-DIV	HRSS MIN-VAR	HRP-VOL	HRP-VAR	MAX BETS PCA COV	MAX BETS PCA COR	MIN-TORSION-COV	MIN-TORSION-COR	ARP	RP-HC	ENSEMBLE
ENSEMBLE															1
RP-HC														1	0.84
ARP													1	0.92	0.91
MIN-TORSION-COR												1	0.74	0.68	0.86
MIN-TORSION-COV											1	0.8	0.96	0.91	0.96
MAX BETS PCA COR										1	0.83	0.84	0.76	0.73	0.87
MAX BETS PCA COV									1	0.8	0.78	0.8	0.71	0.73	0.82
HRP-VAR								1	0.8	0.61	0.57	0.65	0.51	0.59	0.62
HRP-VOL							1	0.5	0.69	0.73	0.95	0.7	0.96	0.93	0.86
HRSS MIN-VAR						1	0.76	0.78	0.88	0.79	0.82	0.76	0.78	0.85	0.83
HRSS MAX-DIV					1	0.84	0.93	0.54	0.75	0.8	0.97	0.77	0.95	0.91	0.94
ERC				1	0.98	0.82	0.96	0.56	0.76	0.81	0.98	0.79	0.97	0.93	0.94
MAX-DIV			1	0.92	0.92	0.77	0.82	0.55	0.77	0.84	0.93	0.83	0.88	0.79	0.97
INV-VAR		1	0.59	0.66	0.64	0.89	0.62	0.93	0.83	0.65	0.66	0.65	0.62	0.72	0.67
INV-VOL	1	0.59	0.76	0.93	0.89	0.73	0.98	0.46	0.65	0.69	0.92	0.65	0.94	0.92	0.81

Source: Data from CSI. Analysis by ReSolve Asset Management.

MEASURING RISK PARITY

While the long-term performance profile of risk parity methods is an important consideration, we must also ask the question: how well is each method achieving the goal of "risk parity"?

To answer this question we need to re-iterate the different objectives described above. Recall that risk parity strategies can have the objective of balancing risk across markets; across asset classes; or across unobservable factors like Principal Components. Ultimately however, the goal is to produce portfolios that are resilient to all of the major economic regimes. We will examine how well each method achieves various risk parity objectives. In addition we will observe the sensitivity of each method to the major economic drivers of growth and inflation. For each objective we are interested in how well each method achieves maximum "balance".

Diversity across individual markets

For our first case study we are interested in measuring how well each method achieves risk balance across all of the individual markets. To measure this we compute the ex ante risk contributions at each rebalance date for each market. Then we calculate the diversity of individual market risk contributions using the inverse Herfindahl index. The inverse Herfindahl index is commonly used to measure the concentration of industries by examining the distribution of market share. The index proxies the number of effective individual actors, so a higher index represents greater diversity.

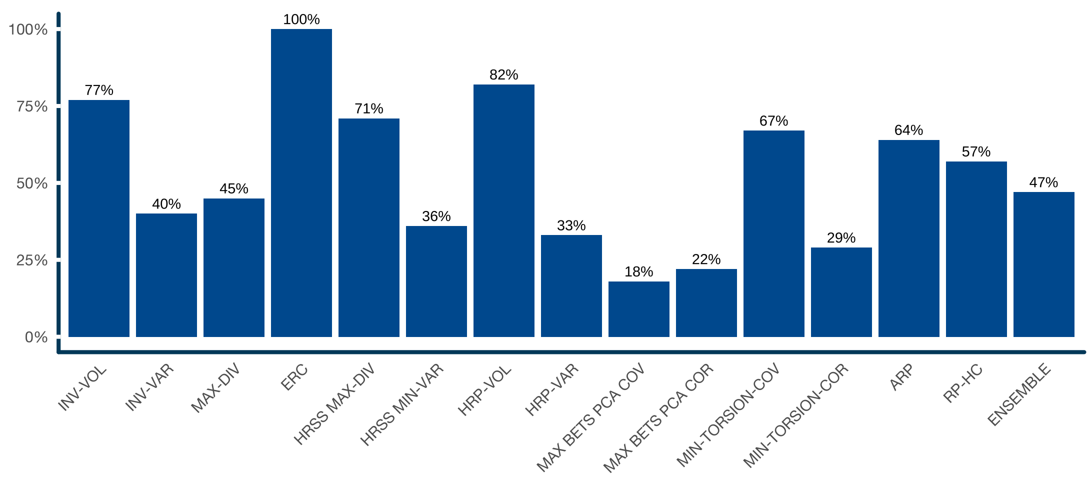
$$RC_i = w_i \times \beta_{i,P} \times \sigma_P$$

$$H^{-1} = \frac{\sum_i^n (RC_i)^2}{{}_i^n \; RC_i^2}$$

Recall that the Equal Risk Contribution (ERC) method has the explicit objective of giving equal risk to each market. This portfolio will always imply 64 effective risk bets, and will therefore score 100% at each rebalance. Other methods will fall short of the ERC portfolio on this metric. For example, if the inverse Herfindahl index of the risk contributions for each market in a risk parity portfolio at time t imply 40 effective bets, and there are 64 markets, 40 bets span 40 / 64 = 62.5% of total possible bets.

We compute effective risk bets as a percent of all available markets at each rebalance date, and then average all of these across time to produce a summary metric between 0 and 1, where 1 implies perfect risk diversity.

Figure 9: **Diversity as percent of total portfolio risk attributable to each market**



Source: Data from CSI. Analysis by ReSolve Asset Management.

Diversity by number of independent bets

The Maximum Diversification method has the goal of maximizing the Diversification Ratio of the portfolio

$$DR = \frac{\sum w \times \sigma}{w \cdot C \cdot w^T}$$

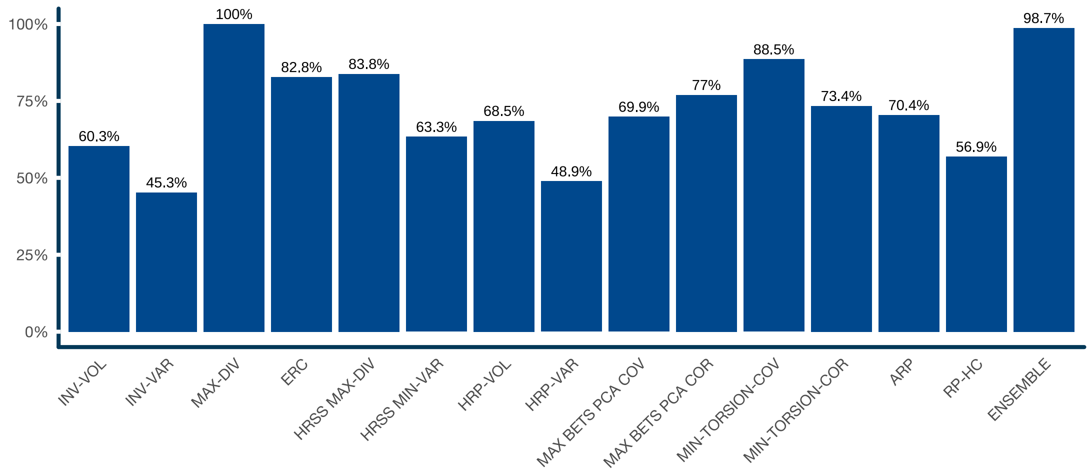
where w is the weight vector, w^T is the transpose of the portfolio weight vector, C is the covariance matrix of returns, and σ is the vector of standard deviations of asset returns.

Choueifaty[9] showed that the number of independent risk factors in a portfolio can be expressed as the square of the Diversification Ratio. By definition, the Maximum Diversification portfolio expresses the maximum possible number of independent bets. All other risk parity method portfolios will express fewer bets.

To measure diversity as a proportion of available independent bets we compute the number of independent bets for each method at each rebalance date and divide by the number of bets expressed by the Maximum Diversification portfolio. This results in a number between 0 and 1, where the Maximum Diversification portfolio will always have a score of 1 by definition. Lastly, we take the average of this score over all rebalance dates.

[9] See Properties of the Most Diversified Portfolio

Figure 10: **Diversity as percent of total possible independent bets as measured by Diversification Ratio^2**



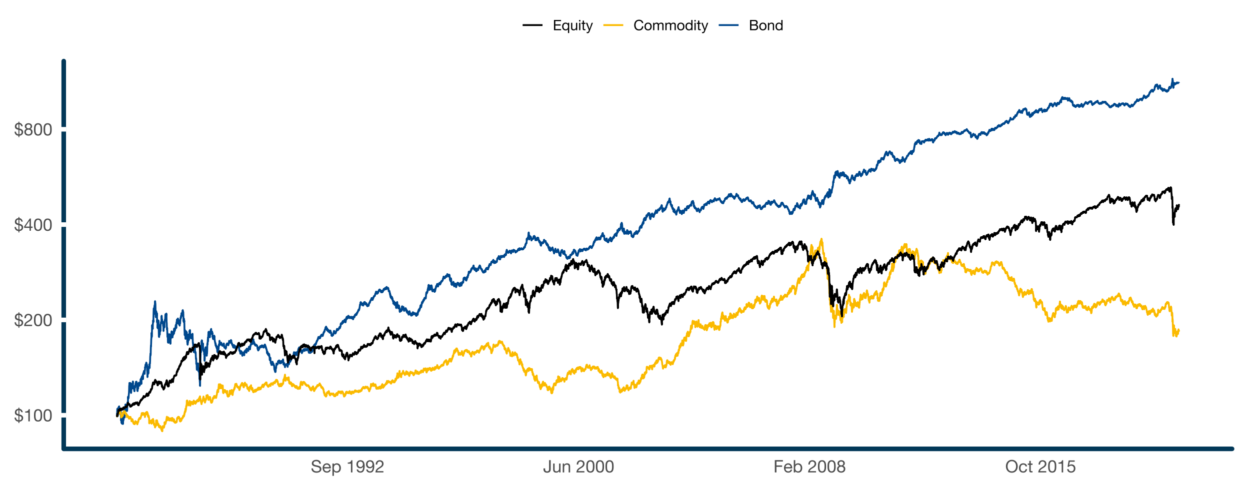
Source: Data from CSI. Analysis by ReSolve Asset Management.

Diversity by asset class

Our investment universe consists of 64 markets divided between 21 equity indices, 13 bond indices, and 30 commodity markets. A classical view of risk parity would seek to gain equal exposure to each of these major clusters.

We create asset class clusters by forming portfolios of assets in each cluster and weighting them at each rebalance date by inverse variance. We use inverse variance because our a priori assumption is that markets in each cluster should exhibit similar volatilities and pairwise correlations, and equal expected returns.

Figure 11: **Compound growth of asset class clusters scaled to 10 percent ex-post volatility, 1985 - May 2020 - SIMULATED RESULTS**



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Source: Data from CSI. Analysis by ReSolve Asset Management. SIMULATED RESULTS.

Observe that the bond cluster produced the best risk-adjusted performance over the in-sample period, with a Sharpe ratio of 0.71 vs equity and commodity cluster Sharpe ratios of 0.47 and 0.22, respectively. It was obviously preferable to overweight bonds in-sample.

Taking our asset class clusters as de facto synthetic assets, at each rebalance date we compute the total volatility attributable to each asset class cluster for each risk parity portfolio. We observe the contemporaneous betas derived from the interaction between the cluster covariance matrix, the market covariance matrix, and the weight vector.

$$F = f \cdot C \cdot w$$

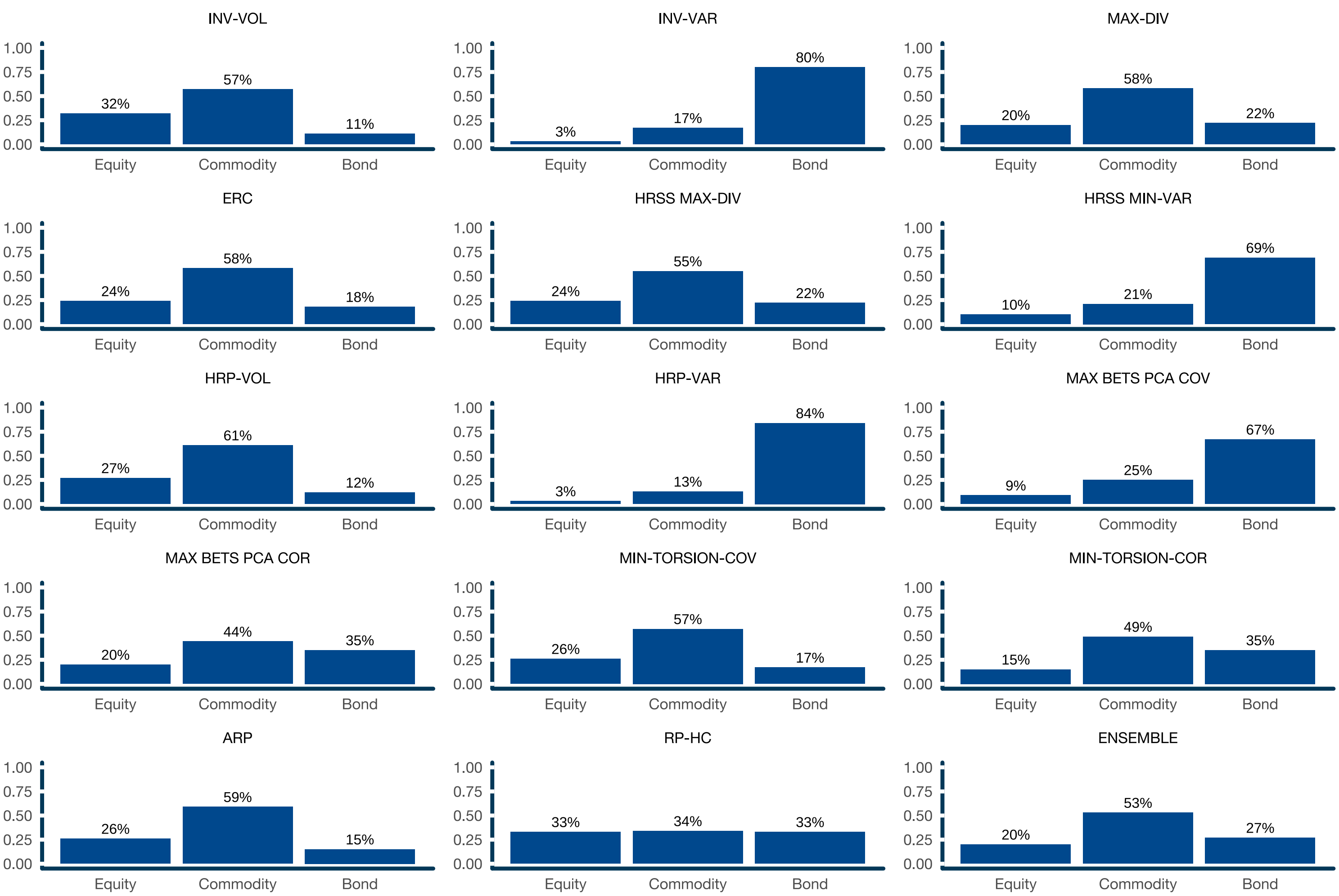
$$\sigma_P^2 = w \cdot C \cdot w^T$$

$$\beta_f = F / \sigma_P^2$$

$$RC_f = \frac{w_f \times \beta_f}{\sum w_f \times \beta_f}$$

where F is the covariance matrix of asset class clusters, f is a 3×64 matrix of weights for each asset class cluster, w is the risk parity portfolio weight vector, C is the market covariance matrix, β_f is the vector of betas of the risk parity portfolio to each asset class cluster, w_f is the cumulative portfolio weight in each asset cluster, and RC_f is the proportion of total risk attributable to each asset class cluster.

Figure 12: **Percent of total average risk contribution attributable to Equity, Bond and Commodity clusters for each method, 1985 - May 2020**

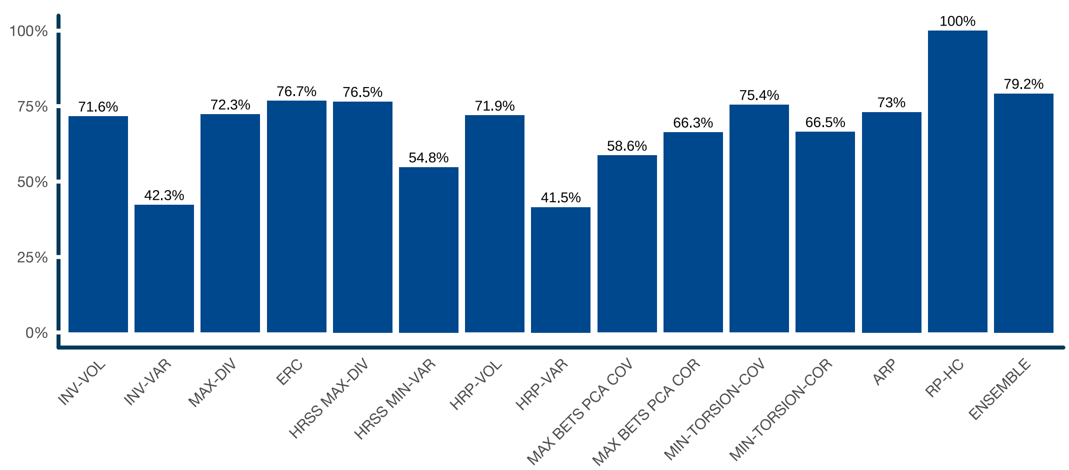


Source: Data from CSI. Analysis by ReSolve Asset Management.

It's worth pausing to examine the distribution of risk contribution by asset class for the different methods. Inverse variance based methods like INV-VAR, HRSS MIN-VAR, and HRP-VAR load heavily on Bond risk. Given that bonds have by far the highest Sharpe ratio in our examination period this bias would be expected to result in higher in-sample return to bond-biased methods. However, a large bias toward bonds implies a large bias toward deflationary regimes, and commensurate vulnerability to inflationary shocks. In addition, given that bond yields are very near all-time historical lows around the world at the time of writing, a strategic bias toward bonds may suggest future returns to bond-heavy methods should be much lower going forward than what is implied by our simulations.

Finally, we calculate the inverse Herfindahl ratio of the asset class risk contributions to determine how evenly risk is divided between the three asset classes. This results in a number between zero and 1, which we then average across all rebalance dates.

Figure 13: **Diversity as percent of total portfolio risk attributable to asset class clusters (i.e. equity, bond, and commodity clusters)**



Source: Data from CSI. Analysis by ReSolve Asset Management.

We explored the concept of measuring exposures to principal factors but since we employ different methods and modifications we deemed the analysis to be unhelpful.

Diversity across economic environments

While the measures of diversity or "parity" above offer important context about the composition and character of risk parity portfolios formed using different methods, ultimately investors are concerned with whether their risk parity portfolio will prove resilient to all of the major economic regimes that might unfold in the future.

It's important to understand that markets are constantly adjusting prices to reflect investor expectations about the future. As a result, meaningful changes in prices will only occur if investors receive new information that is inconsistent with current expectations. When this happens, investors experience a shock, which causes them to adjust the price of assets higher or lower to reflect this new reality.

To make this concept more concrete, imagine that investors are currently expecting a poor environment for a certain asset. To reflect these pessimistic expectations, investors will have acted accordingly to lower the price of the asset. If the future environment is unfavorable for the asset, the price of the asset should not change. That's because investors have already priced the asset appropriately for an unfavorable future. The price of the asset will only be reset higher or lower if investors receive new information that causes a meaningful change in expectations.

To summarize this critical point, asset prices do not change in response to favorable or unfavorable environments. Rather, asset prices reflect investors' current expectations about a favorable or unfavorable environment. Prices will only experience meaningful change if investors receive new information that represents a shock to their current expectations.

The primary drivers of risk for investment portfolios are changes in expectations about future inflation or growth[10]. Economic regimes can be described as combinations of shocks in the directions of inflation or growth, leading to four potential scenarios: inflationary growth, disinflationary growth, inflationary stagnation, and deflationary stagnation.

We want to examine how well our risk parity methods respond to each of these four economic regimes. This is actually quite challenging. While different sources publish inflation and growth expectations data there is considerable disagreement on the optimal model. We chose a simple proxy based on the observation that investors tend to extrapolate the recent past in setting expectations. Thus we measure changes in inflation expectations as the difference between 12-month change in CPI and 36-month change in CPI. We measure changes in growth expectations by measuring the difference between 12-month change in GDP and 36-month change in GDP. We standardized both CPI and GDP series to quarterly frequency.

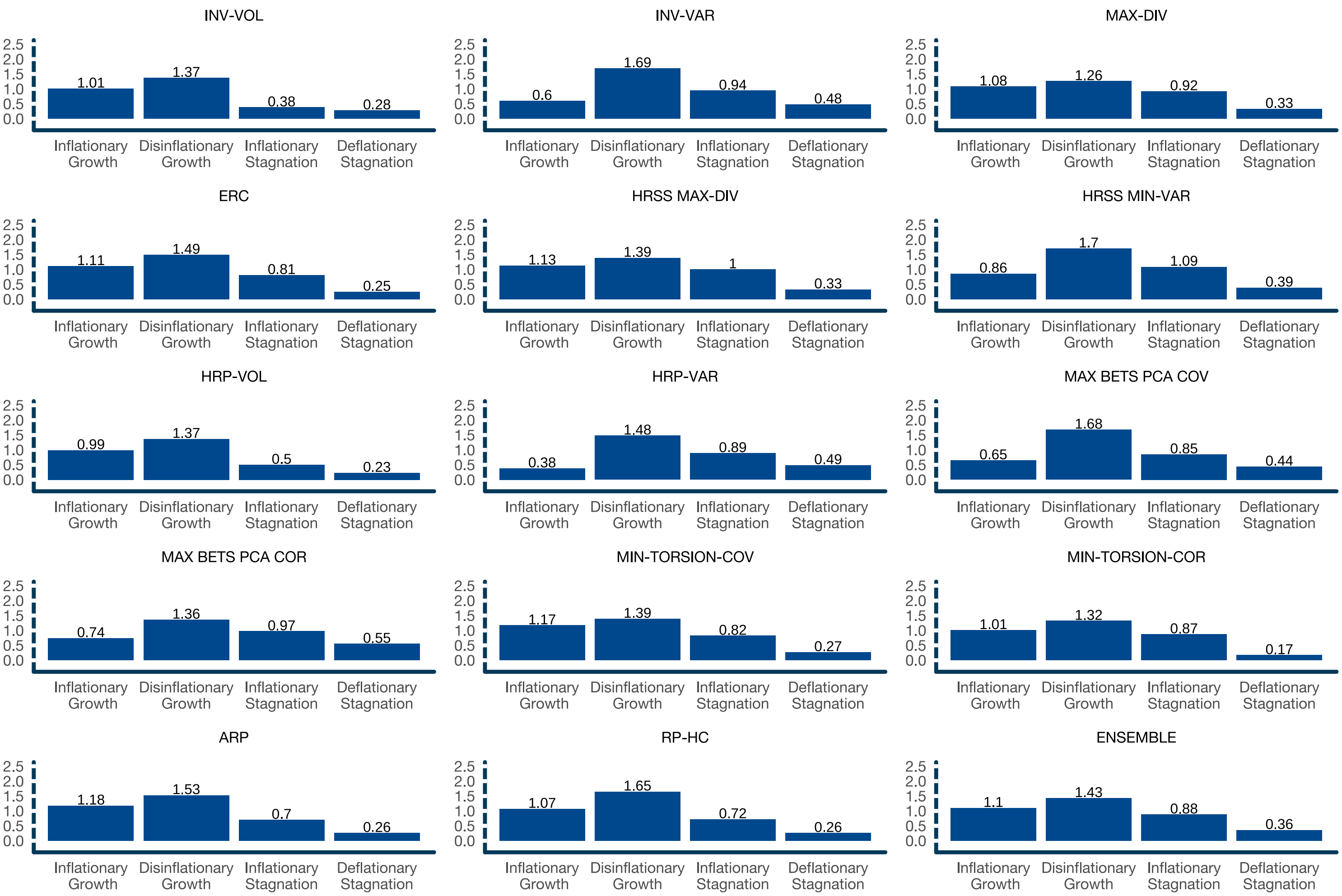
Commensurately, we measured the Sharpe ratios for each of the risk parity methods during quarterly periods corresponding to each of the four regimes as defined below:

- Inflationary growth: Positive inflation shocks and positive growth shocks
- Disinflationary growth: Negative inflation shocks and positive growth shocks
- Inflationary stagnation: Positive inflation shocks and negative growth shocks
- Deflationary stagnation: Negative inflation shocks and negative growth shocks

Note that the four regimes did not occur with equal frequency, so the average of the Sharpe ratios across regimes will not equate to the long-term Sharpe ratio of the strategy.

[10] For a more comprehensive treatment of economic risks and risk parity please see Risk Parity and the Four Faces of Risk

Figure 14: **Average Sharpe ratios by strategy conditioned on economic regime - SIMULATED RESULTS**



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Source: Data from CSI. Analysis by ReSolve Asset Management. SIMULATED RESULTS.

Our hope as risk parity investors is that we will experience a similar investment experience across all four economic regimes. However, it’s clear that many methods are biased toward certain regimes.

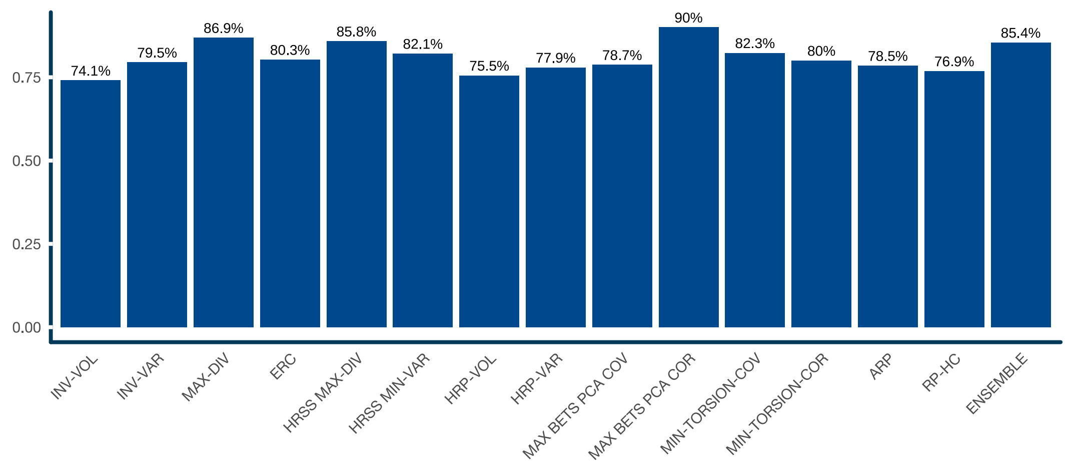
Almost all risk parity methods exhibited a preference for disinflationary growth regimes in our sample. This may reflect the fact that the examination period was biased toward this regime. However, we also observe that variance-based methods like INV-VAR, HRSS MIN-VAR and HRP-VAR, and MAX BETS PCA COV[11] are especially predisposed to this regime. Inverse volatility oriented methods like INV-VOL, ERC, and HRP-VOL all exhibit more balanced results.

Perhaps surprisingly RP-HC, which divides risk by asset class clusters, was more sensitive to economic regimes. In contrast and of particular note, methods that ignore the variance dimension when identifying directions of risk, such as MAX-DIV, HRSS MAX-DIV, MAX BETS PCA COR and MIN-TORSION-COR all produced admirable performance across all four regimes. The ENSEMBLE also exhibited balanced exposure across regimes.

[11] Principal Component Decomposition of the covariance matrix groups assets by both variance and correlation, which both favour bonds

It is illustrative to score each method on the "parity" of their Sharpe ratios across the four regimes using the inverse Herfindahl metric. The MAX-DIV, MAX BETS PCA COR and ENSEMBLE methods stand out as providing the most stable performance across macro-economic environments.

Figure 15: **Diversity as measured by risk-adjusted returns attributable to different economic regimes**



Source: Data from CSI. Analysis by ReSolve Asset Management.

Extensions

So far we've explored the risk parity concept through the traditional lens of asset allocation. However, the concept of risk parity can be expanded to apply to any universe of markets, strategies, or other sources of risk and return.

The concept of diversification can extend to other sources of return that behave differently from the major asset classes themselves. These returns derive from structural and/or behavioural inefficiencies that can be exploited by profit-seeking investors.

For example, investors are often slow to adapt to new information, and are prone to extrapolate intermediate-term trends[12], while underestimating the tendency of corporate or economic prospects to revert to the mean[13]. Many investors also express a preference for "lottery ticket"-type investments; eschew leverage; prefer investments with positive skew; avoid taking short positions[14]; and chase performance at precisely the wrong horizon[15]. They also overpay for insurance against large short-term losses[16]. In the commodity sector producers are motivated to lock-in forward prices to hedge their business risk, and speculators can earn a risk premium for taking on this risk. Other systematic profit opportunities arise due to structural factors or the actions of global institutions that may pursue non-wealth-maximizing objectives to achieve political or longer-term policy goals[17].

These and other risk premia and behavioural/structural inefficiencies leave a variety of profit opportunities for skilled, unconstrained investors to exploit. They manifest as "style premia", "factors" or "edges" that produce returns derived from different sources at different times. In other words, they add to our bench of uncorrelated strategies in pursuit of the most efficient portfolio.
Our focus is on so-called "multi-asset" profit opportunities that manifest across global stock and bond indexes, commodities and currencies. Multi-asset anomalies arise from the same forces as securities-based anomalies, but often have larger barriers to arbitrage, which suggests the anomalies may be larger and more persistent than those observed in the cross-section of individual securities.

[12] See "Time-Series Momentum" by Moskowitz et al.

[13] See "Mean Reversion in Stock Prices: Evidence and Applications" by Poterba and Summers

[14] See "Betting Against Beta" by Frazzini and Pedersen

[15] See Institutionalizing Countercyclical Investment: A Framework for Long-term Asset Owners by Bradley Jones

[16] See "Asset Allocation Implications of the Global Volatility Premium"

[17] See "Carry" by Koijen et al.

At the multi-asset level, most institutions and private clients of major firms have their asset allocation range set by a policy committee, guided by long-term capital market expectations. Institutions rarely take on material active risk at this level of the portfolio. Some institutions are bound by rigid actuarial rules and can only tolerate very small deviations from policy weights. Committee level decisions are typically slow, incremental, and reactive[18]. And peer-oriented compensation schemes heavily and asymmetrically penalize short-term tracking-error to the detriment of long-term alpha generation. These are large and persistent barriers to arbitrage that suggest multi-asset anomalies may exhibit a long shelf-life.

A plain-vanilla risk parity portfolio produces the highest return per unit of risk (i.e. is mean-variance optimal) when markets are expected to produce excess return in proportion to their risk. While this may be a good place to start, global macro market inefficiencies provide opportunities to enhance a risk parity strategy by taking active positions informed by systematic macro-anomalies.

We consider six systematic strategies informed by some of the macro-inefficiences identified above and expressed on the same investment universe as we employed for our risk parity analysis:

- Trend - Take long(short) positions in futures markets with positive(negative) trends over a variety of lookbacks from about 1 month to 12 months
- Carry - Take long(short) positions in futures markets with high(low) carry (i.e. the expected return on a market if the price doesn't change, i.e. from interest rate differentials and/or the expected roll yield from front-month futures to spot, or back-month to front-month futures.)
- Skewness - Take long(short) positions in futures markets with negative(positive) skewness measured in a variety of ways over many lookback horizons
- Seasonality - Take long(short) positions in futures markets with persistent positive(negative) returns at certain times of the year
- Value - Take long(short) positions in futures markets with carry scores that are above(below) their long-term average value
- Volatility - Take long(short) positions in VIX and VSTOXX volatility futures (deferred contract) when historical volatility is greater than(less than) implied volatility; spot term structure is in backwardation(contango); and spot term structure curvature is convex(concave)

With the exception of volatility, all portfolios are traded daily but weights are smoothed to create a tranching effect with a centre-of-mass rebalance frequency of about 5 days (i.e. weekly). At each rebalance, portfolios are formed by resampling all dimensions of portfolio parameters, and subset resampling of the investment universe. Each sample is fed into a proprietary mean-variance optimizer and the final portfolio is the simple average of all sample portfolios.

Figure 16: **Systematic strategy performance, June 1990 - May 2020 - SIMULATED RESULTS**

	Annualized Return	Annualized Volatility	Sharpe Ratio	Max Drawdown	R Ulcer
Trend	7.25%	9.80%	0.76	-18.30%	0.98
Carry	9.61%	8.30%	1.15	-13.10%	2.09
Skewness	6.96%	9.30%	0.77	-14.80%	1.34
Seasonality	14.42%	9.20%	1.50	-15.30%	3.52
Value	8.80%	9.00%	0.98	-17.60%	1.42
Volatility	48.95%	37.00%	1.25	-42.80%	3.02

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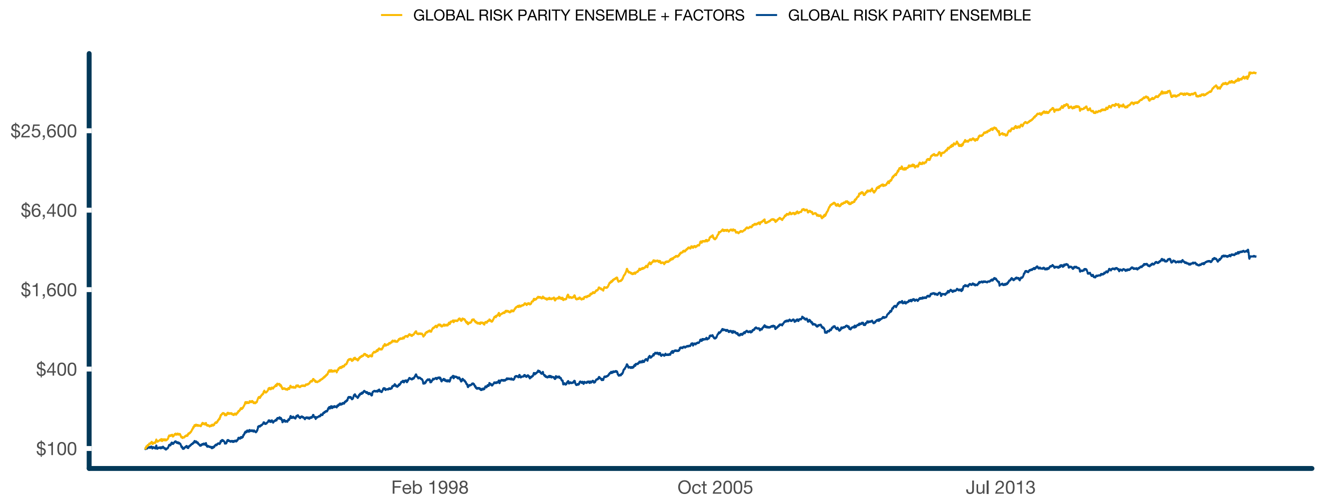
Source: Data from CSI. Analysis by ReSolve Asset Management. SIMULATED RESULTS.

Our Extended Risk Parity strategy derives 50 percent of risk from the Ensemble risk parity strategy described above, and 50 percent from an equal risk allocation across all six systematic strategies. Note that the risk parity portfolio typically expresses between six and eight independent directions of risk and we are adding six generally uncorrelated alternative strategies, so this method approximates a risk parity portfolio across all conventional and alternative risk/return sources.

[18] See Institutionalizing Countercyclical Investment: A Framework for Long-term Asset Owners by Bradley Jones

We scale the final portfolio to target 10 percent annualized volatility based on 20-day estimated covariances as above.

Figure 17: **Compound growth of Extended Risk Parity vs. Ensemble Risk Parity, June 1990 - May 2020 - SIMULATED RESULTS**



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Source: Data from CSI. Analysis by ReSolve Asset Management. SIMULATED RESULTS.

Figure 18: **Historical GROSS performance of Extended Risk Parity vs. Ensemble Risk Parity, June 1990 - May 2020 - SIMULATED RESULTS**

	Start Date	Annualized Return	Volatility	Sharpe Ratio	Max Drawdown	Return-to-Ulcer Ratio
GLOBAL RISK PARITY ENSEMBLE + FACTORS	1990-06-15	23.61%	9.60%	2.25	-15.10%	6.21
GLOBAL RISK PARITY ENSEMBLE	1990-06-15	11.44%	9.70%	1.16	-24.80%	1.43

THESE RESULTS ARE BASED ON SIMULATED OR HYPOTHETICAL PERFORMANCE RESULTS THAT HAVE CERTAIN INHERENT LIMITATIONS. UNLIKE THE RESULTS SHOWN IN AN ACTUAL PERFORMANCE RECORD,THESE RESULTS DO NOT REPRESENT ACTUAL TRADING. ALSO, BECAUSE THESE TRADES HAVE NOT ACTUALLY BEEN EXECUTED, THESE RESULTS MAY HAVE UNDER-OR OVER-COMPENSATED FOR THE IMPACT, IF ANY, OF CERTAIN MARKET FACTORS, SUCH AS LACK OF LIQUIDITY. SIMULATED OR HYPOTHETICAL TRADING PROGRAMS IN GENERAL ARE ALSO SUBJECT TO THE FACT THAT THEY ARE DESIGNED WITH THE BENEFIT OF HINDSIGHT. NO REPRESENTATION IS BEING MADE THAT ANY ACCOUNT WILL OR IS LIKELY TO ACHIEVE PROFITS OR LOSSES SIMILAR TO THESE BEING SHOWN.

Source: Data from CSI. Analysis by ReSolve Asset Management. SIMULATED RESULTS.

There are obviously a mosaic of other directions one might take to improve on the concept including the use of higher-frequency data and more robust methods of volatility/covariance estimation (i.e. HAR), regime labeling (i.e. via volatility and interest rate term structures and curvatures, absorption ratio, turbulence ratio, log-periodic criticality, etc.), and macro-economic relationships. We leave these and other directions for readers to explore.

CONCLUSION

Risk parity strategies are about preparation rather than prediction. The goal is to build a portfolio that would be expected to thrive across any sustained macro-economic environment conditioned on any combination of unexpected inflation and growth. The key ingredients to such a portfolio are diversity and balance. Diversity is about allocating across a universe of investments that respond to inflation and growth shocks in different directions and for different reasons. Balance is about designing portfolios so that low risk and high-risk assets have an equal opportunity to express their unique properties.

It is commonplace in the field of empirical finance to evaluate the merits of an investment strategy based on the long-term performance of a walk-forward simulation. While this is useful, it is critical to ground our analysis in the fundamental assumptions and objectives that informed the rules of portfolio construction.

In the case of risk parity the goal, as stated above, is to achieve steady investment results across economic environments. We often proxy this objective by targeting equal exposure to markets, asset classes or directions of risk. We demonstrated that different risk parity methods exhibit trade-offs between different measures of diversity. Of particular note, we observed that inverse-variance based methods like INV-VAR, HRP-VAR and HRSS-MIN-VAR exhibit a large systematic bias toward bonds in the sample period. Given that bonds produced a much higher Sharpe ratio than equities or commodities it is not surprising that these methods achieved the highest in-sample performance.

However, this highlights a major challenge with using in-sample performance to judge how a strategy is likely to perform out of sample. This is because the in-sample period may not be representative of the distribution of future economic regimes. The examination period was dominated by deflationary periods, and bond yields fell from very high levels near the start of the period to very low levels today. These dynamics produced a strong tail-wind for bonds in-sample. However, risk parity portfolios should be agnostic to inflation and growth surprises. Therefore, portfolios with a systematic bias toward bonds are unlikely to achieve this objective.

Thus, investors would do well to evaluate the merits of risk parity strategies in terms of the fundamental objectives, rather than relying on long-term simulated performance. Importantly, this also means investors should not read too much into live out-of-sample performance of different risk parity strategies, since in many cases results will simply reveal that the systematic biases in these strategies happened to align with the economic shocks experienced by markets in the live period.

Lastly, we extended the traditional concept of risk parity by introducing alternative systematic strategies derived from macro-economic risk factors like carry, skewness value, and volatility; and anomalies like trend and seasonality. An equal risk allocation across traditional and alternative risk factors produced a large improvement in long-term performance with the potential to add further dimensions of resilience across economic regimes.

By way of postscript, all investors should understand – deep in their bones – that every asset allocation framework makes certain core assumptions that almost everyone takes for granted. Notably, they assume that risky assets will earn a premium over cash on average over the long-term.

While this has been true for most markets (though not all), and for global markets in aggregate over the past century or more, there is a small but non-zero probability of a – perhaps unimaginable – event that effectively wipes out all investors at once. Thus, there is an implicit “we are all in this together” bargain embedded in any decision to place money in banks or investments rather than burying cash (or gold, or bitcoin wallets) in your backyard.

For this reason, investors who truly want to hedge all future scenarios may want to invest some proportion of assets in strategies that are fundamentally designed to hedge against unimaginable risks. These typically involve option strategies, which means that they are also vulnerable to a collapse of financial markets. Thus, the truly prepared investor shouldn’t skimp on canned goods, defensive gear, and waterproof matches.

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